Earners AI and ML by Aditya Jain sir - Lecture 05

Link to telegram community : https://t.me/EarnersAdityaJain

Recap

```
if 4 > 3:
  print("A")
```

```
A
```

```
if 4 < 2:
  print("A")
```

```
if 4 < 2:
  print("A")
else:
  print("B")
```

```
B
```

```
if 4 > 2:
  print("A")
elif 4 <2:
  print("B")
else:
  print("C")
```

```
A
```

```
if 4 > 2:
  print("A")
elif 1 < 2:
  print("B")
else:
  print("C")
```

```
A
```

Practice Problems:

```
# Write a program to check whether a person is eligible to vote.

# If age is 18 or above, print "Eligible to Vote", otherwise print "Not Eligible to Vote".
```

```
age = int(input())

if age >= 18:
  print("Eligible to Vote")
else:
  print("Not Eligible to Vote")
```

```
18
Eligible to Vote
```

```
age = int(input())

if age >= 18:
  print("Eligible to Vote")
else:
  print("Not Eligible to Vote")
```

```
20
Eligible to Vote
```

```
age = int(input())

if age >= 18:
  print("Eligible to Vote")
else:
  print("Not Eligible to Vote")
```

```
10
Not Eligible to Vote
```

Q2. Take a number from the user and check whether it is even or odd.

```
n = int(input())

if n%2 == 0:
  print("Even")
else:
  print("Odd")
```

```
5
Odd
```

```
n = int(input())

if n%2 == 0:
  print("Even")
else:
  print("Odd")
```

```
6
Even
```

Q3. Write a program to identify whether a number is positive, negative or zero.

```
n = 5

if n >0:
  print("Positive")
elif n<0:
  print("Negative")
else:
  print("Zero")
```

```
Positive
```

```
n = -5

if n >0:
  print("Positive")
elif n<0:
  print("Negative")
else:
  print("Zero")
```

```
Negative
```

```
n = 0

if n >0:
  print("Positive")
elif n<0:
  print("Negative")
else:
  print("Zero")
```

```
Zero
```

Q4. Take two numbers and print the larger number.

If both are equal, print "Both numbers are equal."

```
a = int(input())
b = int(input())

if a>b:
  print("Larger number is", a)
elif a<b:
  print("Larger number is", b)
else:
  print("Both numbers are equal")
```

```
4
8
Larger number is 8
```

```
a = int(input())
b = int(input())

if a>b:
  print("Larger number is", a)
elif a<b:
  print("Larger number is", b)
else:
  print("Both numbers are equal")
```

```
8
4
Larger number is 8
```

```
a = int(input())
b = int(input())

if a>b:
  print("Larger number is", a)
elif a<b:
  print("Larger number is", b)
else:
  print("Both numbers are equal")
```

```
5
5
Both numbers are equal
```

```
Q5.

Assign grades based on marks.

90 and above → Grade A
75–89 → Grade B
60–74 → Grade C
40–59 → Grade D
Below 40 → Fail
```

```
marks = 90

if marks >= 90:
  print("A")
elif marks >=75 and marks <=89:
  print("B")
elif marks >=60 and marks <= 74:
  print("C")
elif marks >=40 and marks <= 59:
  print("D")
elif marks <40:
  print("Fail")
```

```
A
```

```
#better code
marks = 90

if marks >= 90:
  print("A")
elif marks >=75:
```

```
  print("B")
elif marks >=60:
  print("C")
elif marks >=40:
  print("D")
elif marks <40:
  print("Fail")
```

```
A
```

```
#better code
marks = 90

if marks >= 90:
  print("A")
elif marks >=75:
  print("B")
elif marks >=60:
  print("C")
elif marks >=40:
  print("D")
else:
  print("Fail")
```

```
Body Mass Index (BMI) is a simple measure that helps determine whether a
person's weight is appropriate for their height.

Write a Python program that takes the BMI value as input from the user
and displays the corresponding weight category using the following rules:

BMI Value   Category
Less than 18.5  Underweight
18.5 to 24.9    Normal Weight
25.0 to 29.9    Overweight
30.0 or above   Obese
```

```
bmi = float(input())

if bmi < 18.5:
  print("Underweight")
elif bmi <=24.9:
  print("Normal Weight")
elif bmi <=29.9:
  print("Overweight")
else:
  print("Obese")
```

```
17.5
Underweight
```

```
bmi = float(input())

if bmi < 18.5:
  print("Underweight")
elif bmi <=24.9:
  print("Normal Weight")
elif bmi <=29.9:
  print("Overweight")
else:
  print("Obese")
```

```
30
Obese
```

```
Write a Python program that takes an integer as input and checks its divisibility.

Display the appropriate message based on the following conditions:

If the number is divisible by both 3 and 5, print:
Divisible by both 3 and 5
If the number is divisible only by 3, print:
Divisible only by 3
If the number is divisible only by 5, print:
Divisible only by 5
```

```
Otherwise, print:
Not divisible by 3 or 5
```

```
n = 3

if n%3 == 0:
  print("Div by 3")
elif n%5 == 0:
  print("Div by 5")
elif (n%3 == 0) and (n%5 == 0):
  print("Div by both 3 and 5")
else:
  print("Not divisible by 3 or 5")
```

```
Div by 3
```

```
n = 5

if n%3 == 0:
  print("Div by 3")
elif n%5 == 0:
  print("Div by 5")
elif (n%3 == 0) and (n%5 == 0):
  print("Div by both 3 and 5")
else:
  print("Not divisible by 3 or 5")
```

```
Div by 5
```

```
#incorrect
n = 15

if n%3 == 0:
  print("Div by 3")
elif n%5 == 0:
  print("Div by 5")
elif (n%3 == 0) and (n%5 == 0):
  print("Div by both 3 and 5")
else:
  print("Not divisible by 3 or 5")
```

```
Div by 3
```

```
#modified correct code
n = 15

if (n%3 == 0) and (n%5 == 0):
  print("Div by both 3 and 5")
elif n%3 == 0:
  print("Div by 3")
elif n%5 == 0:
  print("Div by 5")
else:
  print("Not divisible by 3 or 5")
```

```
Div by both 3 and 5
```

```
#modified correct code
n = 5

if (n%3 == 0) and (n%5 == 0):
  print("Div by both 3 and 5")
elif n%3 == 0:
  print("Div by 3")
elif n%5 == 0:
  print("Div by 5")
else:
  print("Not divisible by 3 or 5")
```

```
Div by 5
```

```
#modified correct code
n = 3
```

```
if (n%3 == 0) and (n%5 == 0):
  print("Div by both 3 and 5")
elif n%3 == 0:
  print("Div by 3")
elif n%5 == 0:
  print("Div by 5")
else:
  print("Not divisible by 3 or 5")
```

```
Div by 3
```

```
n = 15

if n%3 == 0 and n%5 != 0:
  print("Div by 3")
elif n%5 == 0 and n%3 != 0:
  print("Div by 5")
elif (n%3 == 0) and (n%5 == 0):
  print("Div by both 3 and 5")
else:
  print("Not divisible by 3 or 5")
```

```
Div by both 3 and 5
```

Loops

```
#Print "Hi" 5 times on the screen, each time on a new line.

print("Hi")
print("Hi")
print("Hi")
print("Hi")
print("Hi")
```

```
Hi
Hi
Hi
Hi
Hi
```

```
# print all the int from 1 to 10 on the screen, each on new line.

print(1)
print(2)
print(3)
print(4)
print(5)
print(6)
print(7)
print(8)
print(9)
print(10)
```

```
1
2
3
4
5
6
7
8
9
10
```

Quiz

```
# Q1. What will be the output?
x = 15

if x > 20:
    print("A")
```

```
elif x > 10:
    print("B")
else:
    print("C")
```

A. A

B. B

C. C

D. Error

Ans: B

```
x = 15

if x > 20:
    print("A")
elif x > 10:
    print("B")
else:
    print("C")
```

```
B
```

What is the output?

```
age = 18

if age >= 18:
    print("Adult")

print("Done")
```

A.
Adult

B.
Done

C.
Adult
Done

D. Error

Ans: C

```
age = 18

if age >= 18:
    print("Adult")

print("Done")
```

```
Adult
Done
```

What will this code print?

```
marks = 85

if marks >= 90:
    print("A")
elif marks >= 80:
    print("B")
elif marks >= 70:
    print("C")
else:
    print("D")
```

Ans: B

```
marks = 85
```

```
if marks >= 90:
    print("A")
elif marks >= 80:
    print("B")
elif marks >= 70:
    print("C")
else:
    print("D")
```

```
B
```

```
Q4. Which keyword is used when checking multiple conditions?

A. else

B. elseif

C. elif

D. otherwise

Ans: C
```

```
What is the output?
num = -5

if num > 0
    print("Positive")
else:
    print("Negative")

A. Positive

B. Negative

C. Zero

D. Error

Ans: D
```

```
num = -5

if num > 0
    print("Positive")
else:
    print("Negative")
```

```
  File "/tmp/ipykernel_1465/1601355304.py", line 3
    if num > 0
              ^
SyntaxError: expected ':'
```

Next steps: Explain error

```
# What will be printed?
x = 25

if x > 10:
    print("A")
if x > 20:
    print("B")
```

```
A
B
```

Double-click (or enter) to edit

```
Which comparison operator checks equality?
```

```
A. =

B. ==

C. ===

D. !=

Ans: B
```

```
age = 15

if age >= 18:
    print("Adult")
elif age >= 13:
    print("Teen")
else:
    print("Child")
```

```
Teen
```

```
Which of the following is/are the correct syntax? (MSQ)

A.
if x > 10
    print(x)

B.
if (x > 10):
print(x)

C.
if x > 10:
    print(x)

D.
if x > 10 then:
    print(x)


Ans: C
```

HW

```
What is the output?
num = 0

if num:
    print("True")
else:
    print("False")

A. True

B. False

C. 0

D. Error
```

```
What is the output?
password = "python"

if password == "Python":
    print("Correct")
else:
    print("Incorrect")

A. Correct

B. Incorrect
```

```
C. Error

D. Nothing
```

```
Which condition checks whether a number lies between 10 and 20 (inclusive)?

A.

num >= 10 and num <= 20

B.

num > 10 or num < 20

C.

num == 10 and 20

D.

10 <= num >= 20
```

```
What is the output?
marks = 95

if marks >= 40:
    print("Pass")

elif marks >= 90:
    print("Excellent")

A. Excellent

B. Pass

C.

Pass
Excellent

D. Error
```

```
Which logical operator returns True only when both conditions are True?

A. or

B. not

C. and

D. ==
```